Henry Rotunno - GGIS489 Semester Project

In [2]:
```
import pandas as pd
import geopandas as gpd
import matplotlib.pyplot as plt
```

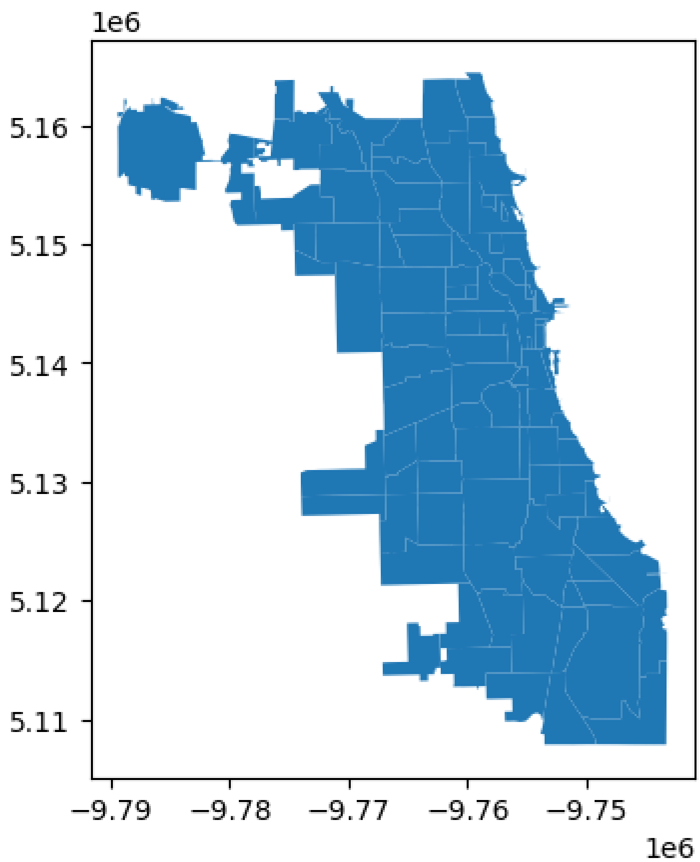
Dataset 1: Chicago neighborhoods

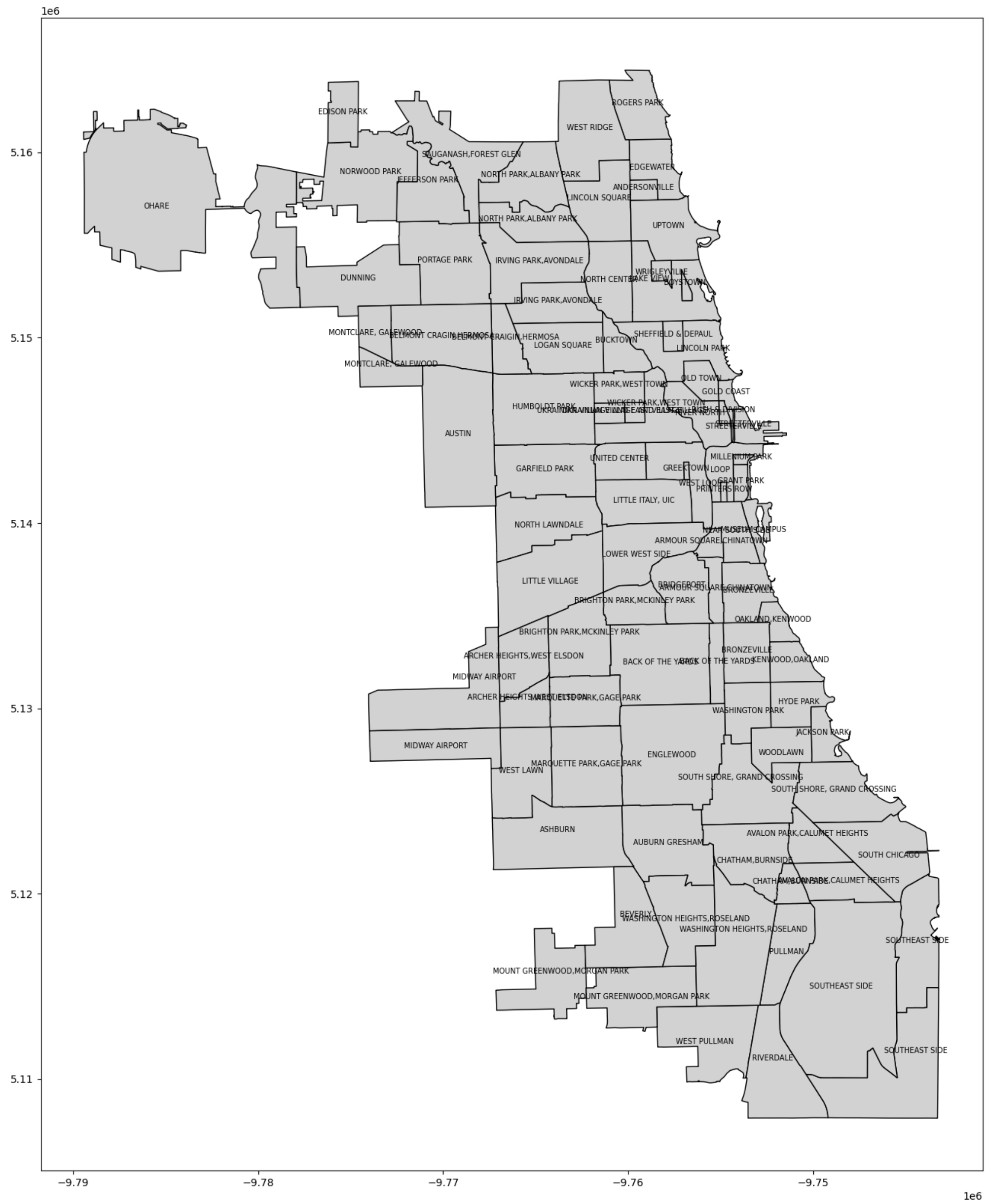
Source: Chicago Data Portal

https://data.cityofchicago.org/Facilities-Geographic-Boundaries/Boundaries-Neighborhoods/bbvz-uum9

In [3]:

```
neighborhoods = gpd.read_file('chi_neighborhoods')
neighborhoods = neighborhoods.to_crs(epsg=3857)
ax = neighborhoods.plot()



fig, ax = plt.subplots(figsize=(30,20))

neighborhoods.plot(ax=ax, color='lightgray', edgecolor='black')


neighborhoods.apply(lambda x:
            ax.annotate(text=x['sec_neigh'],

xy=x.geometry.representative_point().coords[0],
                        ha='center',
                        fontsize=7,
                        ),

                        axis=1)

plt.show()
```





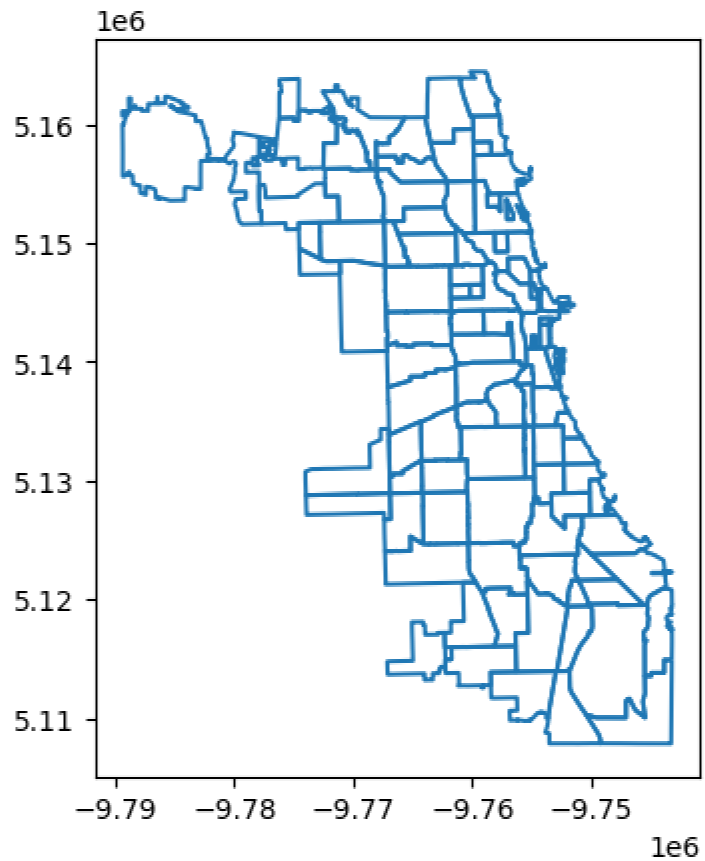
In [4]:
```
# that plot kinda sucks, dont use for presentation
```

In [5]:
```
neighborhoods.columns
```

Out[5]:
```
Index(['pri_neigh', 'sec_neigh', 'shape_area', 'shape_len',
'geometry'], dtype='object')
```

In [6]:
```
neighborhoods.boundary.plot()
```

Out[6]: <Axes: >



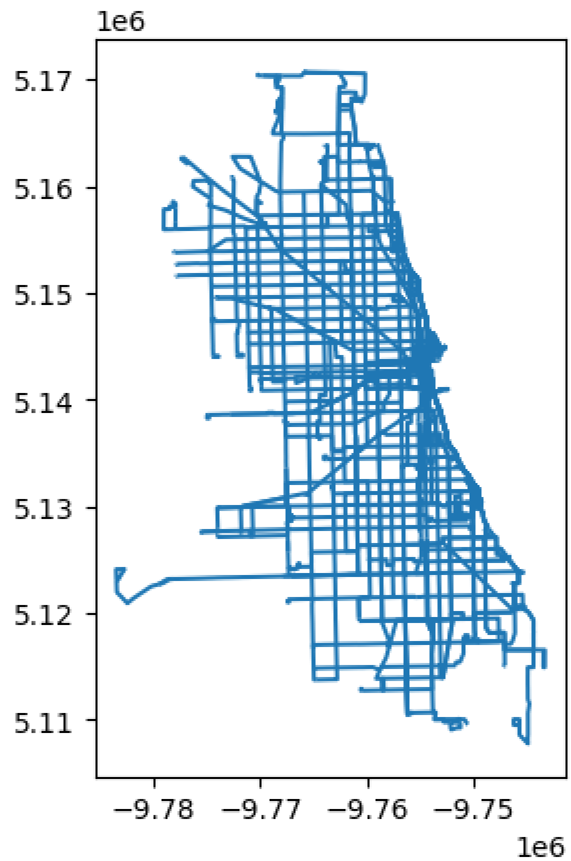
Dataset 2: CTA Bus Routes

Source: Chicago Data Portal

https://data.cityofchicago.org/Transportation/CTA-Bus-Routes-Map/6qfa-9dtu

In [7]:
```
bus_routes = gpd.read_file('bus_routes')

bus_routes = bus_routes.to_crs(epsg=3857)
bus_routes.plot()
```

Out[7]: <Axes: >



In [8]:
```
bus_routes.head()
```

Out[8]:

	route	name	wkday	sat	sun	geometry
0	57	LARAMIE	True	True	True	MULTILINESTRING ((-9768819.651 5141963.938, -9...
1	106	EAST 103RD	True	True	True	MULTILINESTRING ((-9750945.08 5117285.443, -97...
2	192	U OF CHICAGO HOSPITALS EXPRESS	True	False	False	MULTILINESTRING ((-9755955.57 5141114.833, -97...
3	2	HYDE PARK EXPRESS	True	False	False	MULTILINESTRING ((-9767862.303 5123462.161, -9...
4	12	ROOSEVELT	True	True	True	MULTILINESTRING ((-9756920.71 5141096.895, -97...

Dataset 3: CTA Bus Stops

Source: Chicago Data Portal

https://data.cityofchicago.org/Transportation/CTA-Bus-Stops/hvnx-qtky

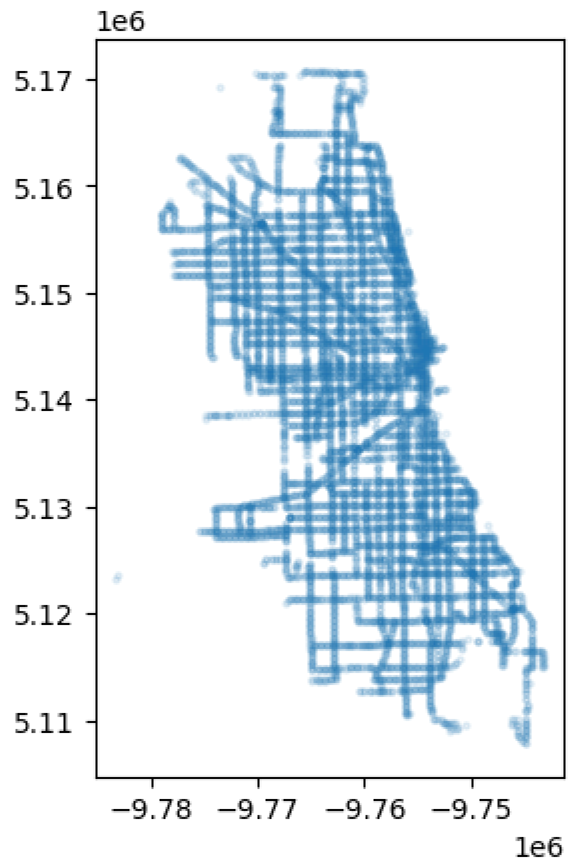
In [9]:
```
bus_stops = gpd.read_file('bus_stops')

bus_stops = bus_stops.to_crs(epsg=3857)

bus_stops.head()
```

Out[9]:

	systemstop	street	cross_st	dir	pos	routesstpg	ow
0	15189.0	CICERO	BERTEAU (north leg)	SB	FS	54,54A	Noı
1	14545.0	HALSTED	119TH STREET	NB	FS	8A,108	Noı
2	15046.0	KEDZIE	TOUHY	NB	FS	11	Noı
3	6280.0	ASHLAND	IRVING PARK	NB	NS	9,X9	Noı
4	4484.0	ARCHER	DAMEN	SWB	NS	62	N6

In [10]:
```
bus_stops.plot(alpha=0.1, markersize=5)
```

Out[10]: <Axes: >



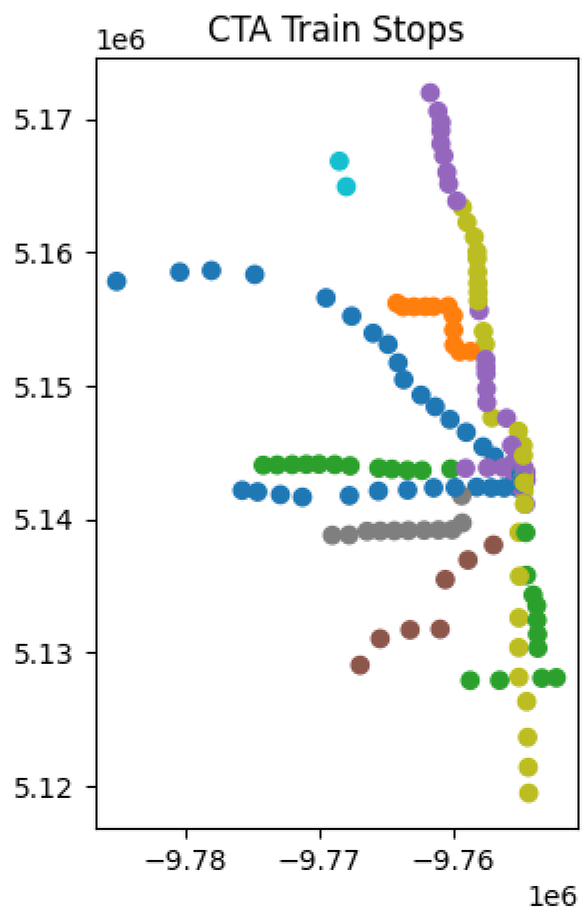
Dataset 4: CTA Rail Stops

Source: Chicago Open Data Portal

https://data.cityofchicago.org/Transportation/CTA-L-Rail-Stations-Map/dytv-z5nm

In [11]:
```
fig, ax = plt.subplots(figsize=(10,5))

l_stops = gpd.read_file('l_stops')
l_stops = l_stops.to_crs(epsg=3857)
ax.set_title('CTA Train Stops')
l_stops.plot(column='legend', ax=ax)
```

Out[11]: <Axes: title={'center': 'CTA Train Stops'}>



Note: The color coordination in these visualizations isn't correct

In [12]:
```
l_stops.head(

)
```

Out[12]:

	station_id	longname	lines	address	ada
0	970.0	Cicero-Congress	Blue Line (Congress)	720 S. Cicero Avenue	False
1	20.0	Harlem-Lake	Green Line (Lake)	1 S. Harlem Avenue	True
2	610.0	Ridgeland	Green Line (Lake)	36 N. Ridgeland Avenue	False
3	230.0	Cumberland	Blue Line	5800 N. Cumberland Avenue	True
4	1700.0	Washington/Wabash	Brown, Orange, Pink, Purple (Express), Green	29 N. Wabash	True

Dataset 5: L Train Routes

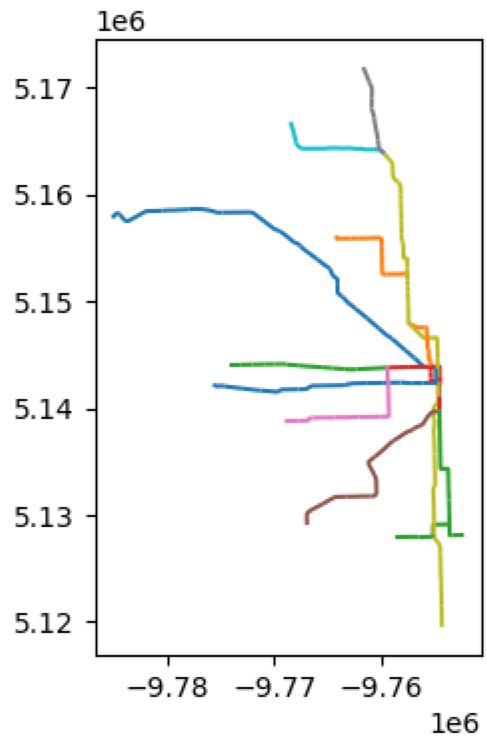
Source: Chicago Data Portal

https://data.cityofchicago.org/Transportation/CTA-L-Rail-Lines-Map/t9su-47fb

In [13]:
```
l_routes = gpd.read_file('l_routes')
l_routes.head()
l_routes = l_routes.to_crs(epsg=3857)
```

In [14]:
```
fig, ax = plt.subplots(figsize=(10,4))
#ax.set_title('CTA Train Routes (L)')
l_routes.plot(column='legend', ax=ax
              )
```

Out[14]: <Axes: >



Dataset 6: ACS Survey

Source: US Census Bureau

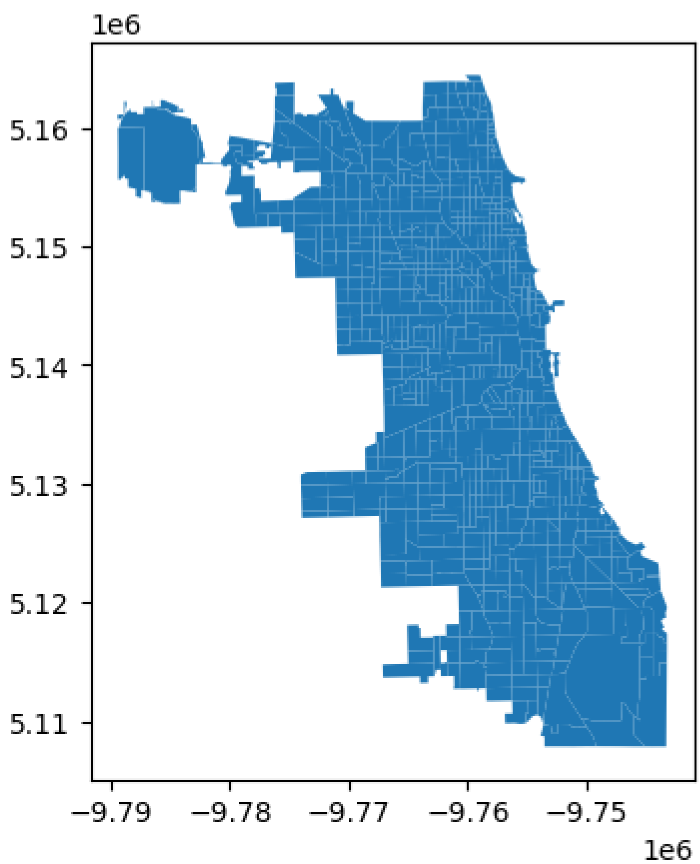
https://www.arcgis.com/home/item.html?id=98c5dd3e1f6247ea8e6e6a88f5660335&sublayer=2&dataTabView=table#data

In [15]:
```
# It took me a while to get this data, I had to tap into some of my
sql knowledge, Im used to just importing it to ArcGIS Pro
url =
'https://services.arcgis.com/GL0fWlNkwysZaKeV/arcgis/rest/services/M
inn_2019_2023_ACS/FeatureServer/2/query?
where=GEOIDFQ%20LIKE%20%271400000US17%25%27&outFields=*&returnGeomet
ry=true&f=geojson'
il_tracts = gpd.read_file(url)
il_tracts = il_tracts.to_crs(epsg=3857)
print((il_tracts.shape))
```

```
(2000, 63)
```

In [16]:
```
chi_tracts = gpd.clip(il_tracts, neighborhoods)

chi_tracts.plot()
```

Out[16]: <Axes: >



In [17]:

```
pd.set_option('display.max_columns', None)

chi_tracts.head(n=5)
```

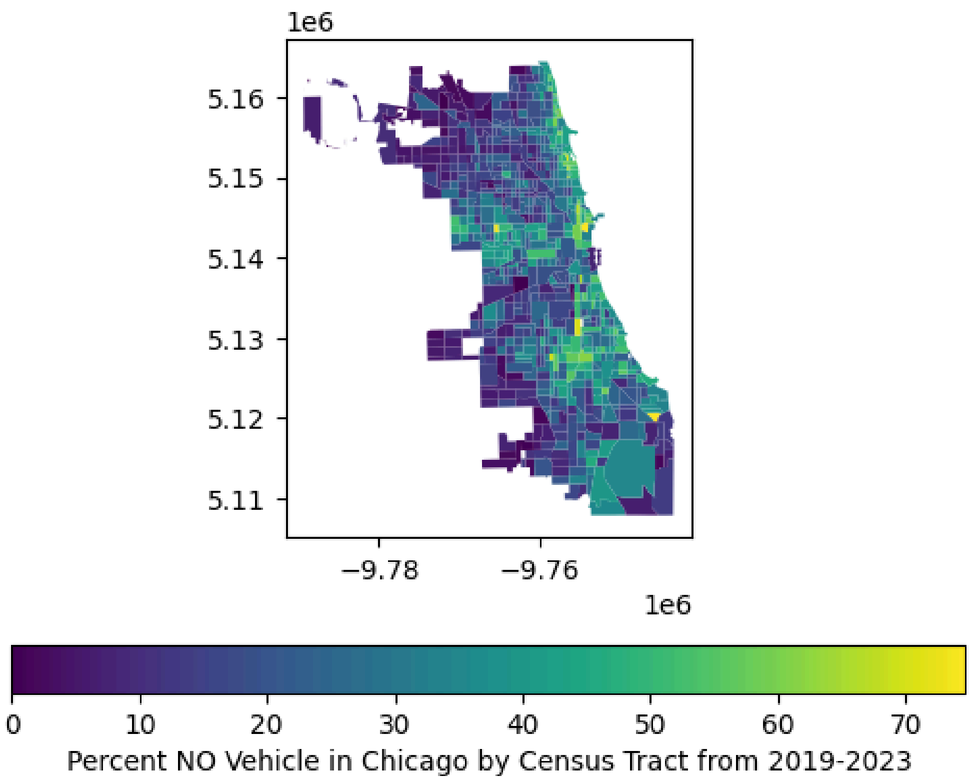
Out[17]:

	OBJECTID	GEOIDFQ	ST	Name	Latitude
727	25419	1400000US17031640700	IL	6407	41.776412
722	25414	1400000US17031640100	IL	6401	41.781929
1480	26172	1400000US17031980100	IL	9801	41.785971
672	25364	1400000US17031560700	IL	5607	41.796467
1409	26101	1400000US17031835200	IL	8352	41.796640

In [18]:

```
chi_tracts.plot(column='Percent_No_Vehicle',
                legend=True,
                legend_kwds={'label': 'Percent NO Vehicle in Chicago
by Census Tract from 2019-2023', 'orientation': 'horizontal'},
                cmap='viridis')
```

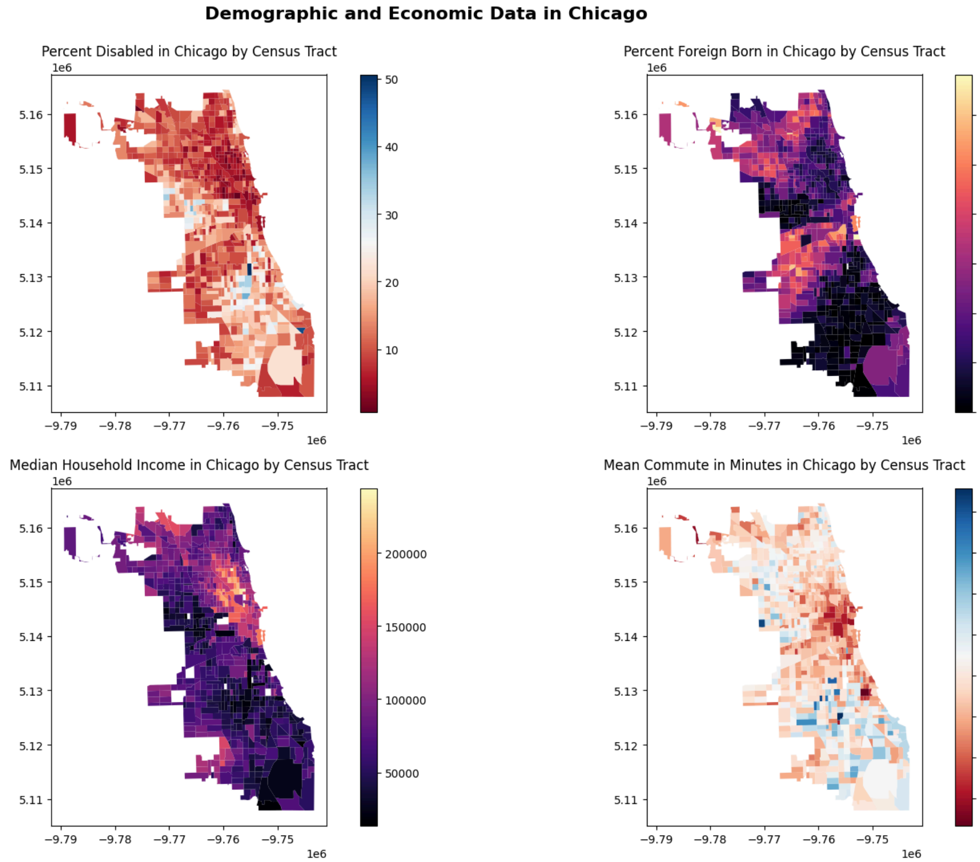
Out[18]: <Axes: >



1: Plotting demographics in Chicago

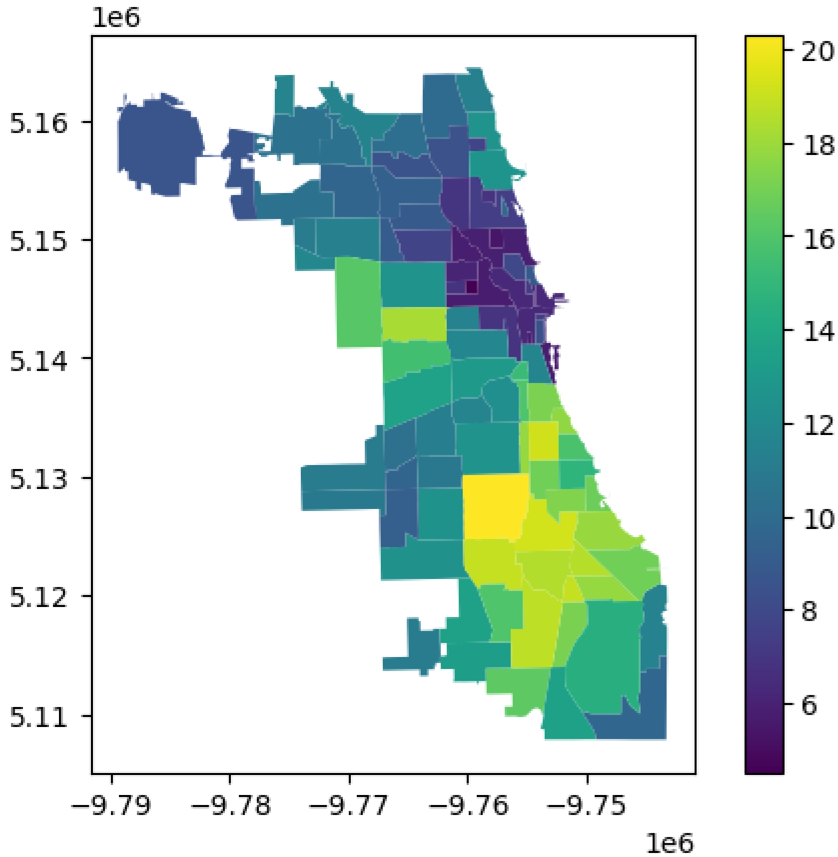
In [19]:

```
fig, axes = plt.subplots(2,2, figsize=(16,11))
chi_tracts.plot(column='Percent_Disabled',
                ax=axes[0,0],
                legend=True,
                cmap='RdBu'
)

chi_tracts.plot(column='Percent_Foreign_Born',
                ax=axes[0,1],
                legend=True,
                cmap='magma')

chi_tracts.plot(column='Median_Household_Income',
                legend=True,
                cmap='magma',
                ax=axes[1,0]
                )

chi_tracts.plot(column='Mean_Commute_Minutes',
                ax=axes[1,1],
                legend=True,
                cmap='RdBu')



axes[0,0].set_title('Percent Disabled in Chicago by Census Tract')
axes[0,1].set_title('Percent Foreign Born in Chicago by Census
Tract')
axes[1,0].set_title('Median Household Income in Chicago by Census
Tract')
axes[1,1].set_title('Mean Commute in Minutes in Chicago by Census
Tract')

plt.tight_layout(rect=[0, 0, 1, 0.95])
plt.suptitle('Demographic and Economic Data in Chicago',
fontsize=16, fontweight='bold')
plt.show()
```


Demographic and Economic Data in Chicago
Percent Disabled in Chicago by Census Tract
Percent Foreign Born in Chicago by Census Tract
Median Household Income in Chicago by Census Tract
Mean Commute in Minutes in Chicago by Census Tract
1e6
5.16
5.15
5.14
5.13
5.12
5.11
−9.79 −9.78 −9.77 −9.76 −9.75
10 20 30 40 50
0 10 20 30 40 50 60
50000 100000 150000 200000
20 25 30 35 40 45 50 55

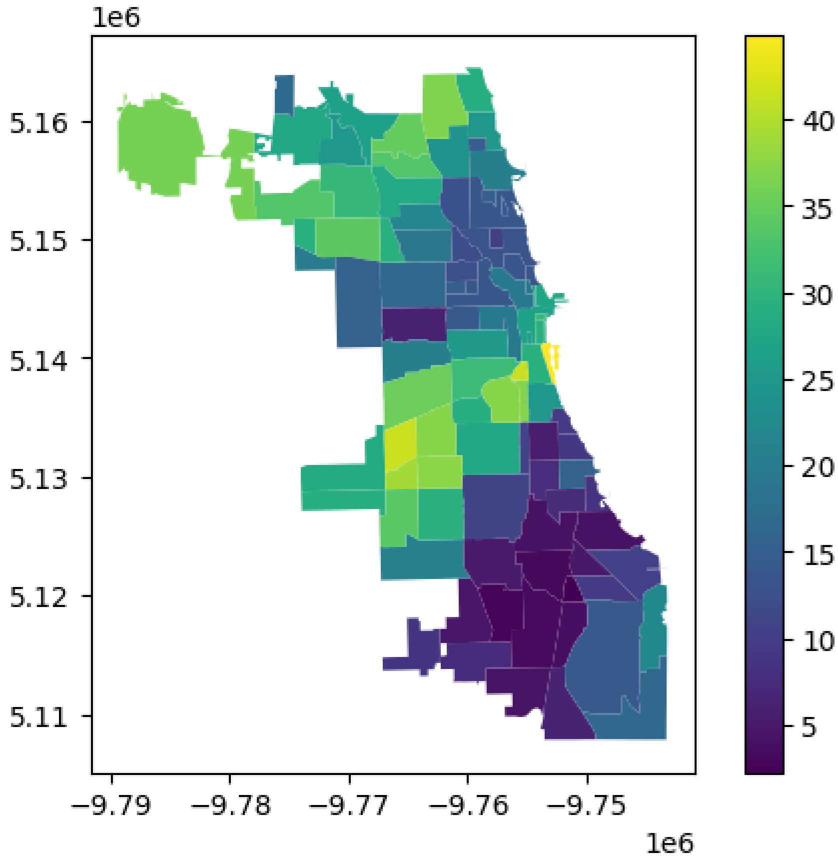
In [20]:
```
neighborhood_tracts = gpd.sjoin(neighborhoods, chi_tracts,
predicate='intersects')



# had to normalize data

neighborhood_tracts['count_disabled'] =
(neighborhood_tracts['Percent_Disabled'] / 100 *
neighborhood_tracts['Total_Population'])



summ = neighborhood_tracts.groupby('pri_neigh').agg(
    total_pop=('Total_Population', 'sum'),
    disabled_count=('count_disabled', 'sum')
)

summ['disabled_rate'] = (
    summ['disabled_count'] / summ['total_pop']
) * 100

merger = neighborhoods.merge(summ, on='pri_neigh')

merger.plot(column='disabled_rate', legend=True)
```

Out[20]: <Axes: >


1e6
5.16
5.15
5.14
5.13
5.12
5.11
−9.79
−9.78
−9.77
−9.76
−9.75
1e6
20
18
16
14
12
10
8
6

In [21]:
```
neighborhood_tracts = gpd.sjoin(neighborhoods, chi_tracts,
predicate='intersects')



# had to aggregate data

#SEE --> https://michaelminn.net/tutorials/arcgis-pro-
proximity/#:~:text=Interstate%20Highways,class%20in%20the%20project%
20geodatabase.

neighborhood_tracts['count_foreign_born'] =
(neighborhood_tracts['Percent_Foreign_Born'] / 100 *
neighborhood_tracts['Total_Population'])



summ_foreign = neighborhood_tracts.groupby('pri_neigh').agg(
    total_pop=('Total_Population', 'sum'),
    foreign_count=('count_foreign_born', 'sum')
)

summ_foreign['foreign_rate'] = (
    summ_foreign['foreign_count'] / summ_foreign['total_pop']
) * 100

merger_foreign = neighborhoods.merge(summ_foreign, on='pri_neigh')

merger_foreign.plot(column='foreign_rate', legend=True)
```

Out[21]: <Axes: >


1e6
5.16
5.15
5.14
5.13
5.12
5.11
−9.79
−9.78
−9.77
−9.76
−9.75
1e6
40
35
30
25
20
15
10
5

In [22]:
```
summ_foreign
```

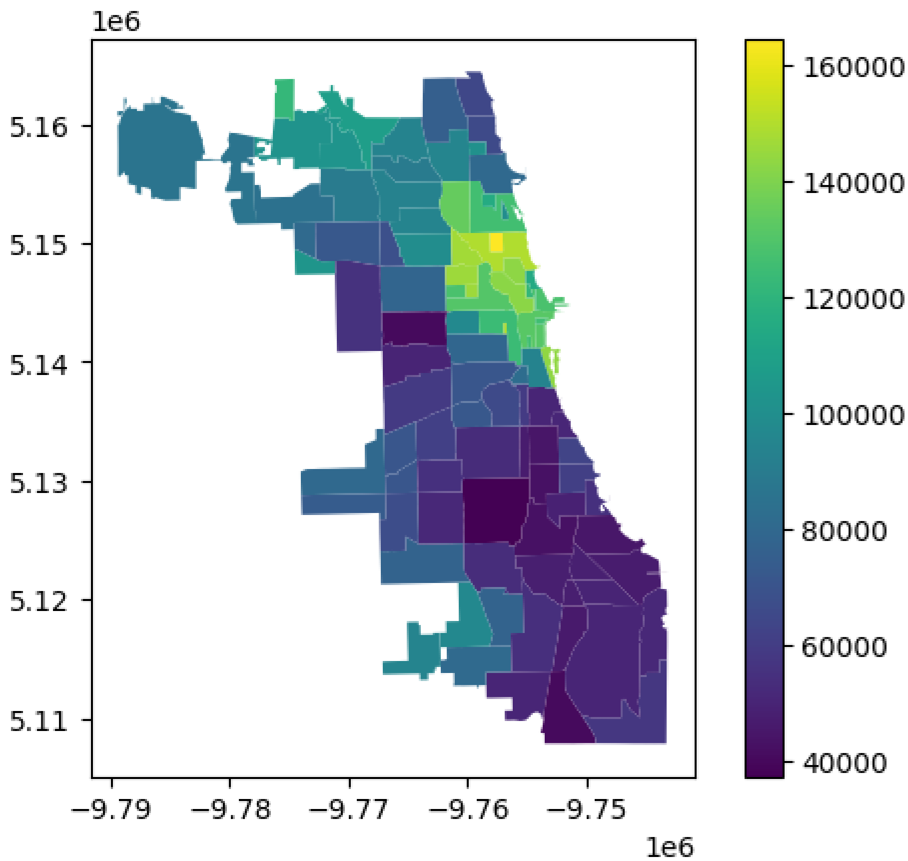
Out[22]:

	total_pop	foreign_count	foreign_rate
pri_neigh			
Albany Park	93026	30977.188	33.299495
Andersonville	29084	4310.526	14.820953
Archer Heights	36942	15265.781	41.323645
Armour Square	29947	11075.915	36.985057
Ashburn	97494	20093.706	20.610198
...	...	...	...
West Ridge	139082	51082.466	36.728308
West Town	100745	13880.845	13.778197
Wicker Park	69852	8729.701	12.497425
Woodlawn	49510	3699.923	7.473082
Wrigleyville	41784	6228.787	14.907110

98 rows × 3 columns

In [23]:
```
median_income = neighborhood_tracts.groupby('pri_neigh')
['Median_Household_Income'].mean()
merger_income = neighborhoods.merge(median_income, on='pri_neigh')
merger_income.plot(column='Median_Household_Income', legend=True)
```

Out[23]: <Axes: >



In [24]:
```
merger_income
```

Out[24]:

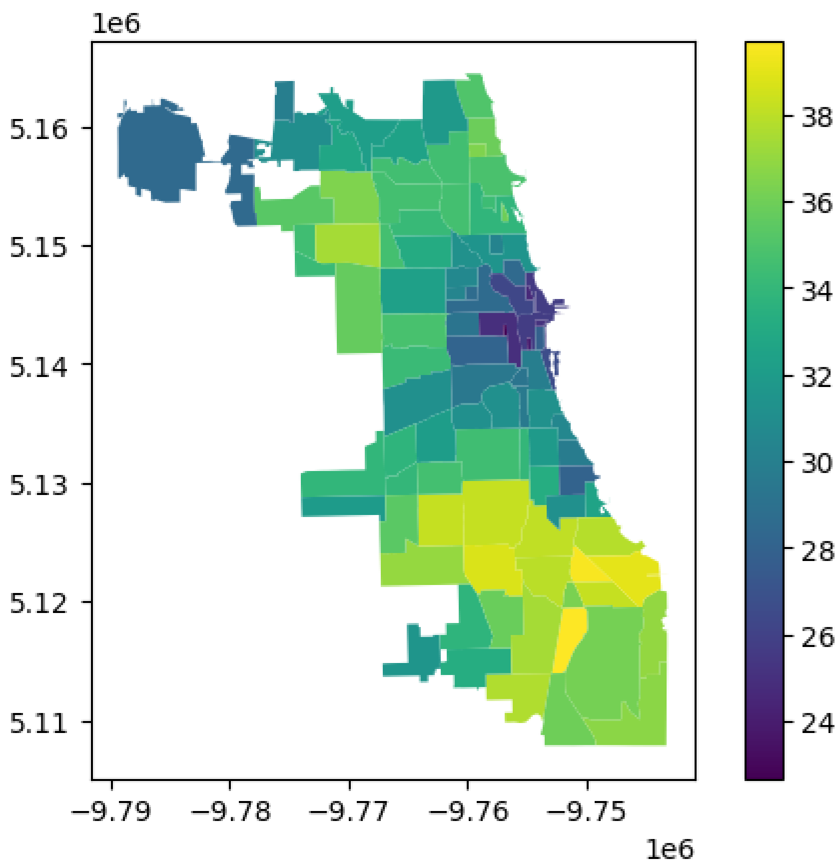
	pri_neigh	sec_neigh	shape_area	shape_len	
0	Grand Boulevard	BRONZEVILLE	4.849250e+07	28196.837157	PO ((- 51 -9
1	Printers Row	PRINTERS ROW	2.162138e+06	6864.247156	PO ((- 51 -9
2	United Center	UNITED CENTER	3.252051e+07	23101.363745	PO ((- 51 -9
3	Sheffield & DePaul	SHEFFIELD & DEPAUL	1.048259e+07	13227.049745	PO ((- 51 -9
4	Humboldt Park	HUMBOLDT PARK	1.250104e+08	46126.751351	PO ((- 51 -9
...	...	...	...	...	...
93	Belmont Cragin	BELMONT CRAGIN,HERMOSA	1.090994e+08	43311.706886	PO ((- 51 -9
94	Austin	AUSTIN	1.700378e+08	55473.345911	PO ((- 51 -9
95	Gold Coast	GOLD COAST	7.165706e+06	13685.479377	PO ((- 51 -9
96	Boystown	BOYSTOWN	3.365779e+06	9780.268985	PO ((- 51 -9

	pri_neigh	sec_neigh	shape_area	shape_len	
97	River North	RIVER NORTH	3.876644e+07	31506.037810	PC ((· 51 -9

98 rows × 6 columns

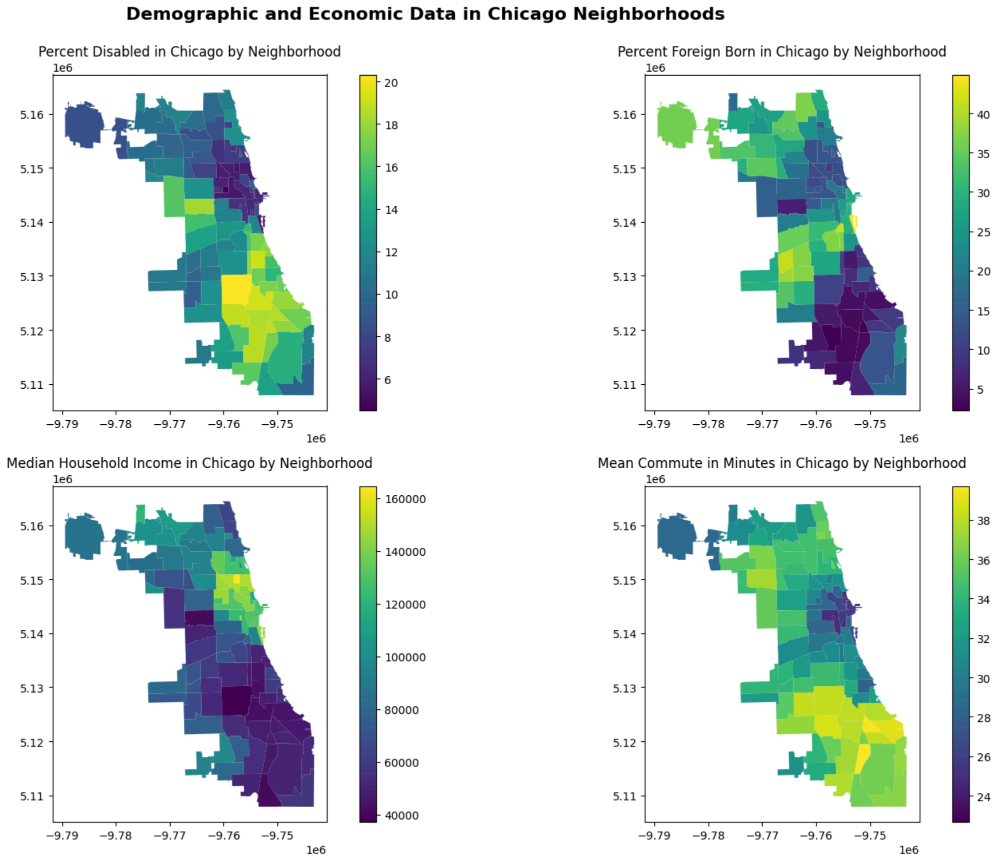
In [25]:
```
commute = neighborhood_tracts.groupby('pri_neigh')
['Mean_Commute_Minutes'].mean()

merger_commute = neighborhoods.merge(commute, on='pri_neigh')
merger_commute.plot('Mean_Commute_Minutes', legend=True)
```

Out[25]: <Axes: >



In [26]:

```
fig, axes = plt.subplots(2,2, figsize=(16,11))

merger.plot(column='disabled_rate',
            ax=axes[0,0],

            legend=True)

merger_foreign.plot(column='foreign_rate',
                    ax=axes[0,1],
                    legend=True)

merger_income.plot(column='Median_Household_Income',
                    ax=axes[1,0],
                    legend=True)

merger_commute.plot(column='Mean_Commute_Minutes',
                    ax=axes[1,1],
                    legend=True)

axes[0,0].set_title('Percent Disabled in Chicago by Neighborhood')
axes[0,1].set_title('Percent Foreign Born in Chicago by
Neighborhood')
axes[1,0].set_title('Median Household Income in Chicago by
Neighborhood')
axes[1,1].set_title('Mean Commute in Minutes in Chicago by
Neighborhood')

plt.tight_layout(rect=[0, 0, 1, 0.95])
plt.suptitle('Demographic and Economic Data in Chicago
Neighborhoods', fontsize=16, fontweight='bold')
plt.show()
```


Demographic and Economic Data in Chicago Neighborhoods
Percent Disabled in Chicago by Neighborhood
Percent Foreign Born in Chicago by Neighborhood
Median Household Income in Chicago by Neighborhood
Mean Commute in Minutes in Chicago by Neighborhood
1e6
5.16
5.15
5.14
5.13
5.12
5.11
−9.79
−9.78
−9.77
−9.76
−9.75
1e6

In [27]:

```
merger_income
```

Out[27]:

	pri_neigh	sec_neigh	shape_area	shape_len	
0	Grand Boulevard	BRONZEVILLE	4.849250e+07	28196.837157	PO ((- 51 -9
1	Printers Row	PRINTERS ROW	2.162138e+06	6864.247156	PO ((- 51 -9
2	United Center	UNITED CENTER	3.252051e+07	23101.363745	PO ((- 51 -9
3	Sheffield & DePaul	SHEFFIELD & DEPAUL	1.048259e+07	13227.049745	PO ((- 51 -9
4	Humboldt Park	HUMBOLDT PARK	1.250104e+08	46126.751351	PO ((- 51 -9
...	...	...	...	...	...
93	Belmont Cragin	BELMONT CRAGIN,HERMOSA	1.090994e+08	43311.706886	PO ((- 51 -9
94	Austin	AUSTIN	1.700378e+08	55473.345911	PO ((- 51 -9
95	Gold Coast	GOLD COAST	7.165706e+06	13685.479377	PO ((- 51 -9
96	Boystown	BOYSTOWN	3.365779e+06	9780.268985	PO ((- 51 -9

	pri_neigh	sec_neigh	shape_area	shape_len	
97	River North	RIVER NORTH	3.876644e+07	31506.037810	PC ((- 51 -9

98 rows × 6 columns

In [28]:
```
merger_commute
```

Out[28]:

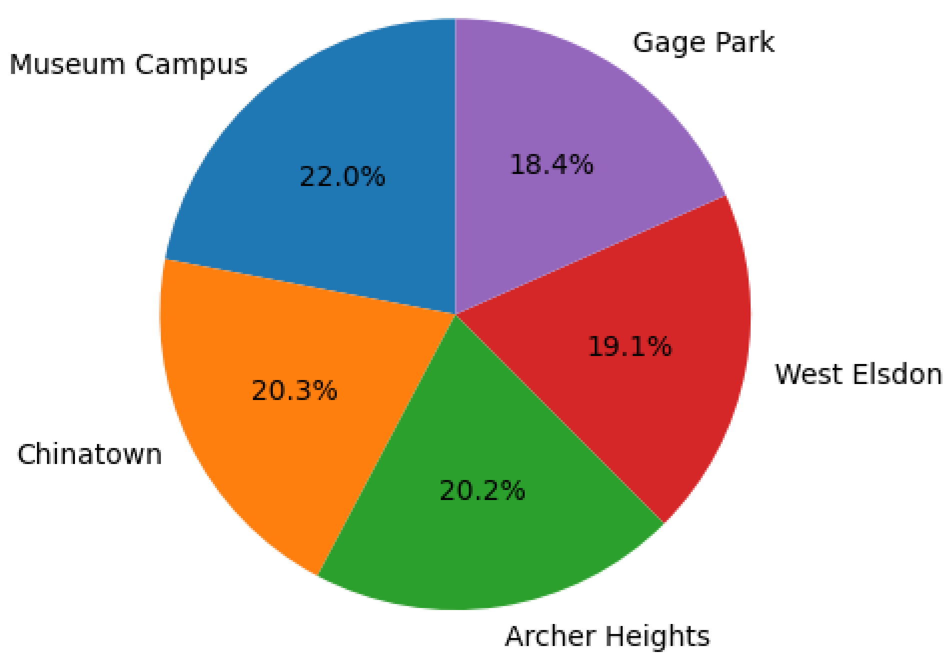
	pri_neigh	sec_neigh	shape_area	shape_len	
0	Grand Boulevard	BRONZEVILLE	4.849250e+07	28196.837157	P(((- 51 -9
1	Printers Row	PRINTERS ROW	2.162138e+06	6864.247156	P(((- 51 -9
2	United Center	UNITED CENTER	3.252051e+07	23101.363745	P(((- 51 -9
3	Sheffield & DePaul	SHEFFIELD & DEPAUL	1.048259e+07	13227.049745	P(((- 51 -9
4	Humboldt Park	HUMBOLDT PARK	1.250104e+08	46126.751351	P(((- 51 -9
...	...	...	...	...	...
93	Belmont Cragin	BELMONT CRAGIN,HERMOSA	1.090994e+08	43311.706886	P(((- 51 -9
94	Austin	AUSTIN	1.700378e+08	55473.345911	P(((- 51 -9
95	Gold Coast	GOLD COAST	7.165706e+06	13685.479377	P(((- 51 -9
96	Boystown	BOYSTOWN	3.365779e+06	9780.268985	P(((- 51 -9

	pri_neigh	sec_neigh	shape_area	shape_len	
97	River North	RIVER NORTH	3.876644e+07	31506.037810	PC ((- 51 -9

98 rows × 6 columns

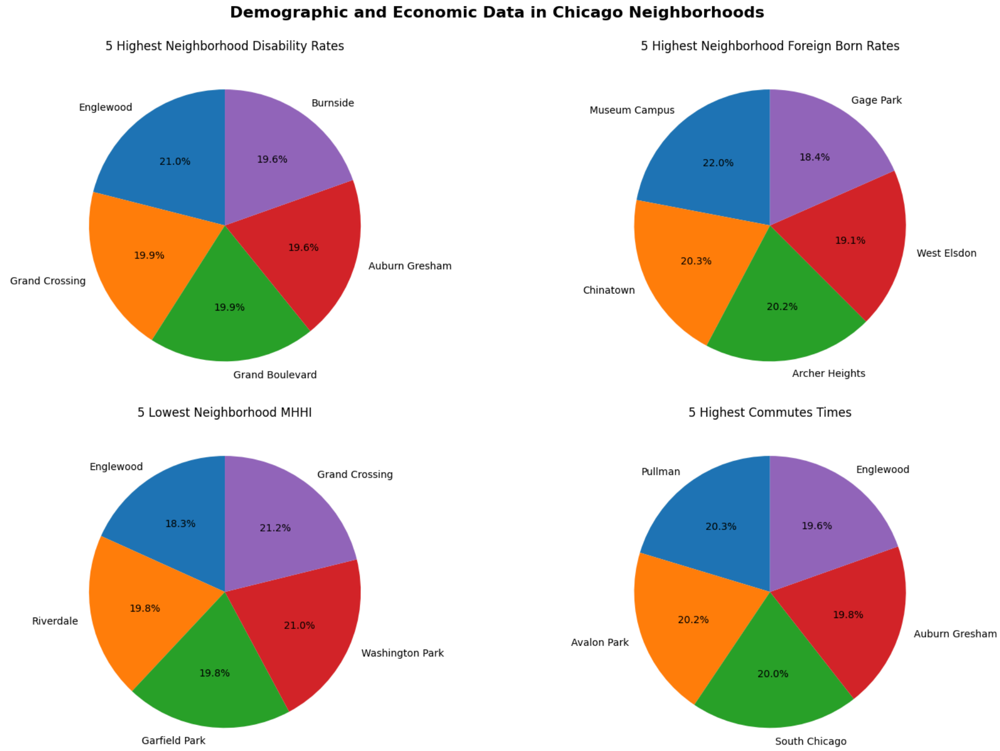
In [29]:
```
bottom_5_dis = merger.nlargest(5, 'disabled_rate')
bottom_5_for = merger_foreign.nlargest(5, 'foreign_rate')

plt.pie(bottom_5_for['foreign_rate'],
labels=bottom_5_for['pri_neigh'], autopct='%1.1f%%', startangle=90)
```

Out[29]:
```
([<matplotlib.patches.Wedge at 0x12d2c8f50>,
  <matplotlib.patches.Wedge at 0x12d330680>,
  <matplotlib.patches.Wedge at 0x12d35e810>,
  <matplotlib.patches.Wedge at 0x12d35ecf0>,
  <matplotlib.patches.Wedge at 0x12d35f770>],
 [Text(-0.7012350647591399, 0.84750774860897, 'Museum Campus'),
  Text(-0.9913802045679949, -0.4766186001308811, 'Chinatown'),
  Text(0.16389266091602733, -1.087722021335352, 'Archer
Heights'),
  Text(1.0811823305792394, -0.20259508395626055, 'West
Elsdon'),
  Text(0.600871308909377, 0.9213868189471414, 'Gage Park')],
 [Text(-0.3824918535049853, 0.46227695378671085, '22.0%'),
  Text(-0.5407528388552699, -0.2599737818895715, '20.3%'),
  Text(0.08939599686328763, -0.593302920728373 8, '20.2%'),
  Text(0.5897358166795851, -0.11050640943068755, '19.1%'),
  Text(0.32774798667784194, 0.5025746285166225, '18.4%')])
```



In [30]:

```
bottom_5_inc = merger_income.nsmallest(5, 'Median_Household_Income')

bottom_5_comm = merger_commute.nlargest(5, 'Mean_Commute_Minutes')

fig, axes = plt.subplots(2,2, figsize=(16,11))

axes[0,0].pie(bottom_5_dis['disabled_rate'],
labels=bottom_5_dis['pri_neigh'], autopct='%1.1f%%', startangle=90,
    )



axes[0,1].pie(bottom_5_for['foreign_rate'],
labels=bottom_5_for['pri_neigh'], autopct='%1.1f%%', startangle=90,
    )



axes[1,0].pie(bottom_5_inc['Median_Household_Income'],
labels=bottom_5_inc['pri_neigh'], autopct='%1.1f%%', startangle=90,
    )

axes[1,1].pie(bottom_5_comm['Mean_Commute_Minutes'],
labels=bottom_5_comm['pri_neigh'], autopct='%1.1f%%', startangle=90,
    )



axes[0,0].set_title('5 Highest Neighborhood Disability Rates')
axes[0,1].set_title('5 Highest Neighborhood Foreign Born Rates')
axes[1,0].set_title('5 Lowest Neighborhood MHHI')
axes[1,1].set_title('5 Highest Commutes Times')

plt.tight_layout(rect=[0, 0, 1, 0.95])
plt.suptitle('Demographic and Economic Data in Chicago
Neighborhoods', fontsize=16, fontweight='bold')
plt.show()
```



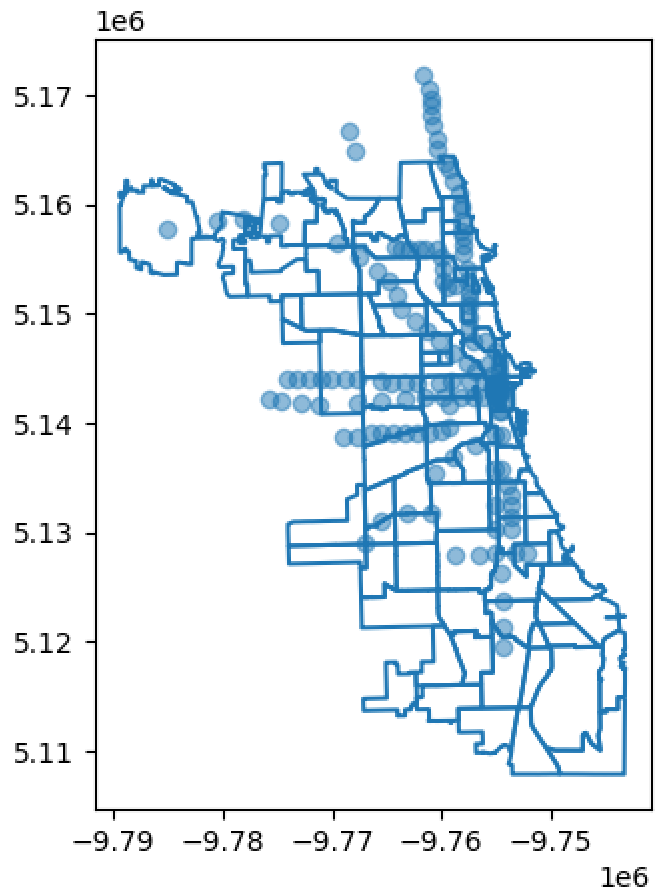
Transportation

Walkability --> within 0.5 miles (800 meters)

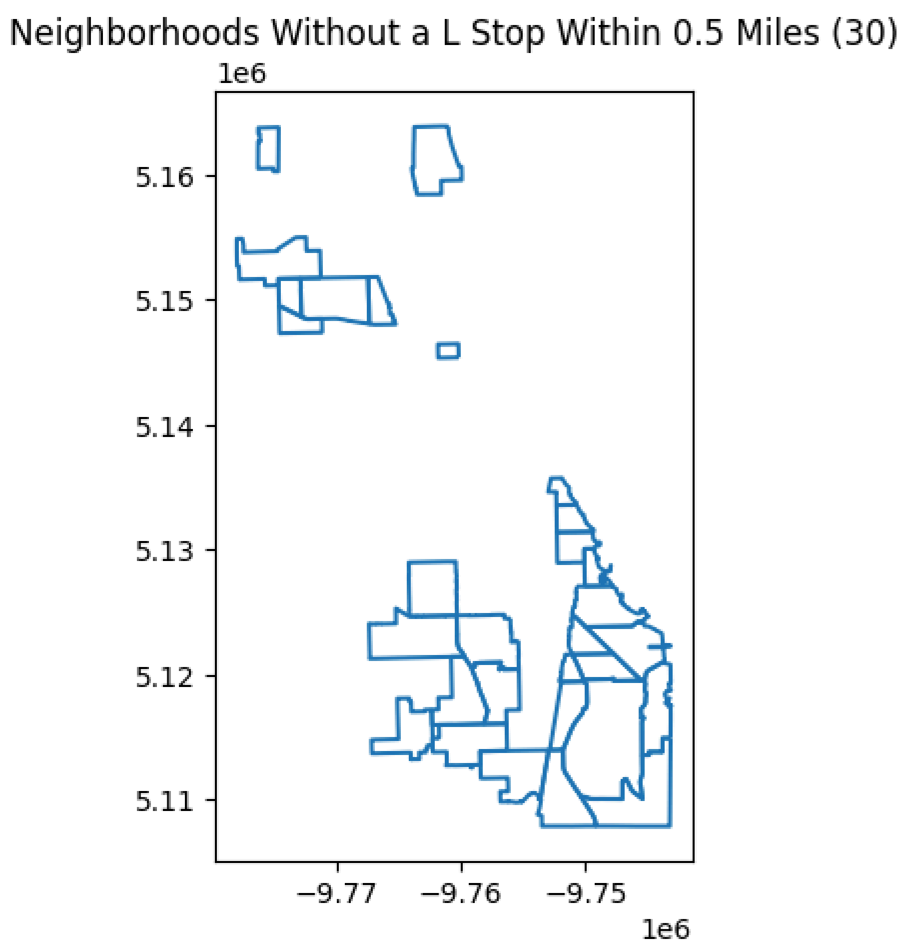
In [31]:

```
fig, ax = plt.subplots(figsize=(10,5))

neighborhoods.boundary.plot(ax=ax)
l_stops.plot(ax=ax, alpha=0.5)
```

Out[31]: <Axes: >



In [ ]:
```
nearest_nodes = gpd.sjoin_nearest(
    neighborhoods,
    l_stops,
    how="left",
    max_distance=800, # --> 0.5 miles
    distance_col="dist_to_node"
)



empty_neighborhoods =
nearest_nodes[nearest_nodes["dist_to_node"].isna()]






fig, ax = plt.subplots(figsize=(8,5))

empty_neighborhoods.boundary.plot(ax=ax)
ax.set_title('Neighborhoods Without a L Stop Within 0.5 Miles (30)')
plt.show()
```



In [ ]:
```
all_together = empty_neighborhoods.merge(merger, on='pri_neigh')
        # join  census data
all_together = all_together.set_geometry("geometry_x")

fig, ax = plt.subplots(figsize=(10,5))
all_together.plot(column='disabled_rate', legend=True, ax=ax)
ax.set_title('Disability Rate of Neighborhoods Without an L Stop
Within 0.5 Miles')
```

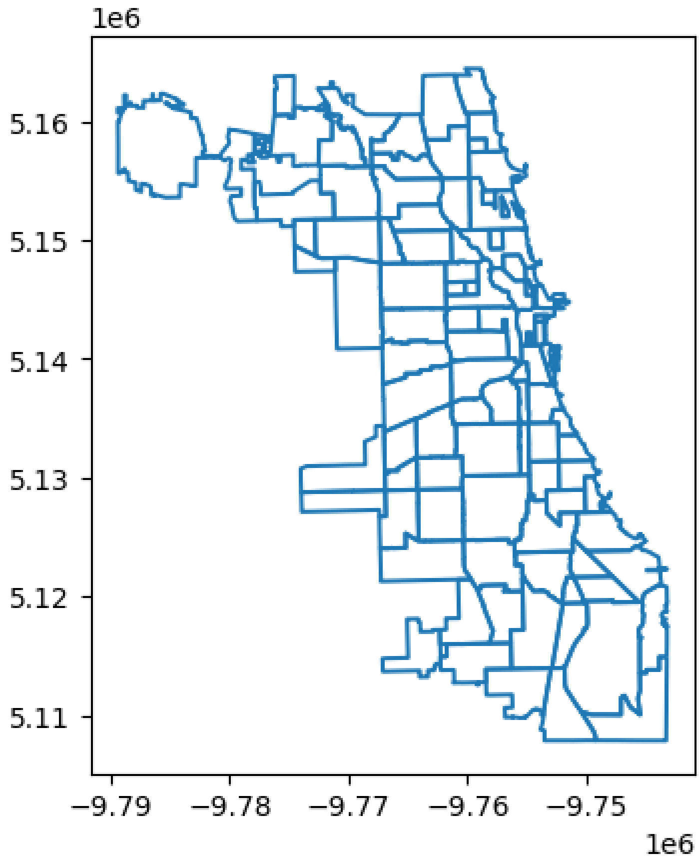
```
Text(0.5, 1.0, 'Disability Rate of Neighborhoods Without an L
Stop Within 0.5 Miles')
```

In [34]:
```
len(empty_neighborhoods)
```

Out[34]: 30

In [35]:
```
neighborhoods.boundary.plot()
```

Out[35]: <Axes: >



In [36]:
```
len(neighborhoods)
```

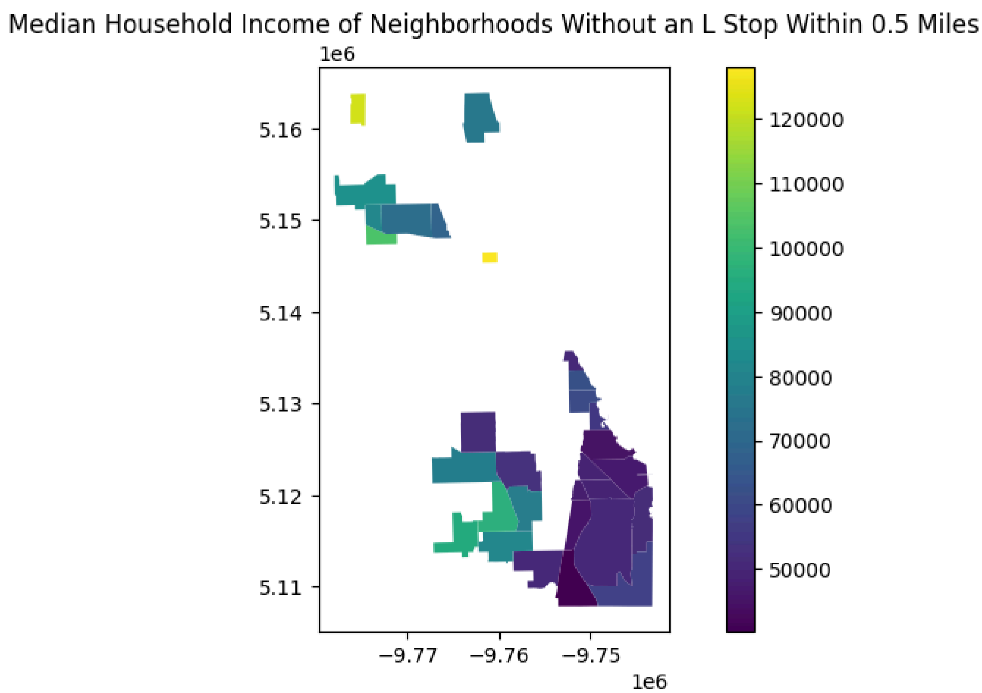
Out[36]: 98

In [37]:
```
30/98
```

Out[37]: 0.30612244897959184

In [38]:
```
all_together = empty_neighborhoods.merge(merger_income,
on='pri_neigh')
all_together = all_together.set_geometry("geometry_x")





fig, ax = plt.subplots(figsize=(10,5))
all_together.plot(column='Median_Household_Income', legend=True,
ax=ax)
ax.set_title('Median Household Income of Neighborhoods Without an L
Stop Within 0.5 Miles')
```

Out[38]:
```
Text(0.5, 1.0, 'Median Household Income of Neighborhoods
Without an L Stop Within 0.5 Miles')
```



In [39]:
```
low = all_together[['pri_neigh', 'Median_Household_Income']].copy()

print(low.nsmallest(7, 'Median_Household_Income'))
```

```
           pri_neigh  Median_Household_Income
11         Riverdale             40219.083333
17       South Shore             44962.666667
10           Pullman             46003.750000
18     South Chicago             46997.750000
1          Burnside              47912.222222
0        Avalon Park             48506.000000
3    Calumet Heights             49826.357143
```

In [40]:

```
merger_foreign
```

Out[40]:

	pri_neigh	sec_neigh	shape_area	shape_len	
0	Grand Boulevard	BRONZEVILLE	4.849250e+07	28196.837157	P(((- 51 -9
1	Printers Row	PRINTERS ROW	2.162138e+06	6864.247156	P(((- 51 -9
2	United Center	UNITED CENTER	3.252051e+07	23101.363745	P(((- 51 -9
3	Sheffield & DePaul	SHEFFIELD & DEPAUL	1.048259e+07	13227.049745	P(((- 51 -9
4	Humboldt Park	HUMBOLDT PARK	1.250104e+08	46126.751351	P(((- 51 -9
...	...	...	...	...	...
93	Belmont Cragin	BELMONT CRAGIN,HERMOSA	1.090994e+08	43311.706886	P(((- 51 -9
94	Austin	AUSTIN	1.700378e+08	55473.345911	P(((- 51 -9
95	Gold Coast	GOLD COAST	7.165706e+06	13685.479377	P(((- 51 -9
96	Boystown	BOYSTOWN	3.365779e+06	9780.268985	P(((- 51 -9

	pri_neigh	sec_neigh	shape_area	shape_len	
97	River North	RIVER NORTH	3.876644e+07	31506.037810	PC ((- 51 -9

98 rows × 8 columns

In [41]:
```
all_together = empty_neighborhoods.merge(merger_foreign,
on='pri_neigh')
all_together = all_together.set_geometry("geometry_x")





fig, ax = plt.subplots(figsize=(10,5))
all_together.plot(column='foreign_rate', legend=True, ax=ax)
ax.set_title('Foreign Born Rate for Neighborhoods Without an L Stop
Within 0.5 Miles')
```

Out[41]:
```
Text(0.5, 1.0, 'Foreign Born Rate for Neighborhoods Without an
L Stop Within 0.5 Miles')
```

In [42]:
```
print(all_together['total_pop'].sum())
```

```
1792859
```

In [43]:
```
all_together.nlargest(5, 'foreign_rate')
```

Out[43]:

	pri_neigh	sec_neigh_x	shape_area_x	shape_len_x	
16	West Ridge	WEST RIDGE	9.842909e+07	43020.689458	F (5 -
29	Belmont Cragin	BELMONT CRAGIN,HERMOSA	1.090994e+08	43311.706886	F (5 -
15	Dunning	DUNNING	1.035950e+08	54555.992534	F (5 -
20	Chicago Lawn	MARQUETTE PARK,GAGE PARK	9.827947e+07	40073.099838	F (5 -
2	Hermosa	BELMONT CRAIGIN,HERMOSA	3.260206e+07	27179.017438	F (5 -

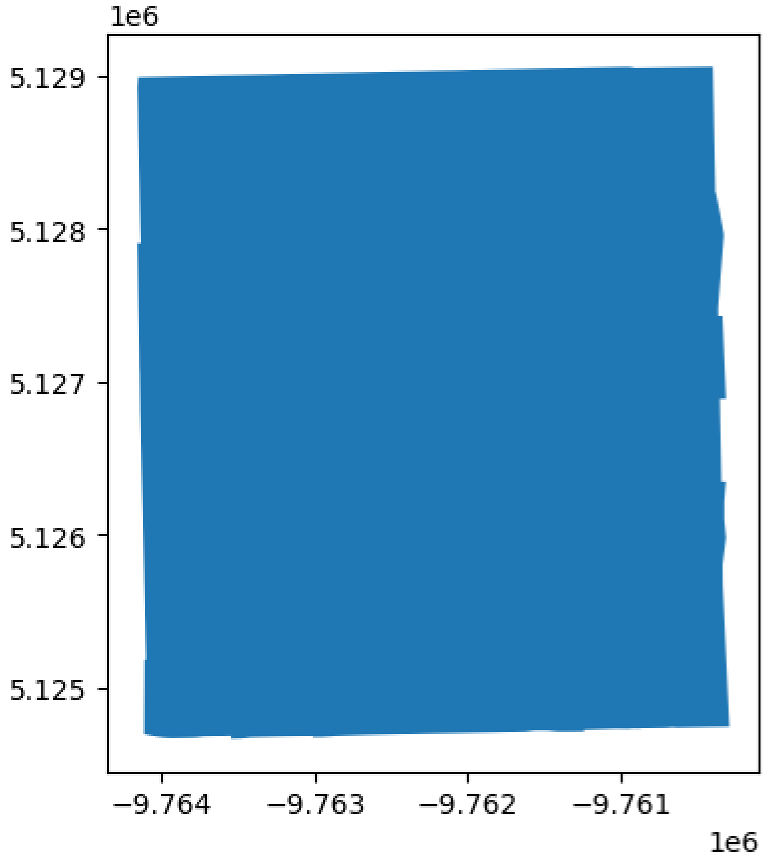
In [44]:
```
high = all_together[['pri_neigh', 'foreign_rate']].copy()
high.nlargest(5, 'foreign_rate')
```

Out[44]:

	pri_neigh	foreign_rate
16	West Ridge	36.728308
29	Belmont Cragin	34.151463
15	Dunning	33.259292
20	Chicago Lawn	29.451339
2	Hermosa	29.406815

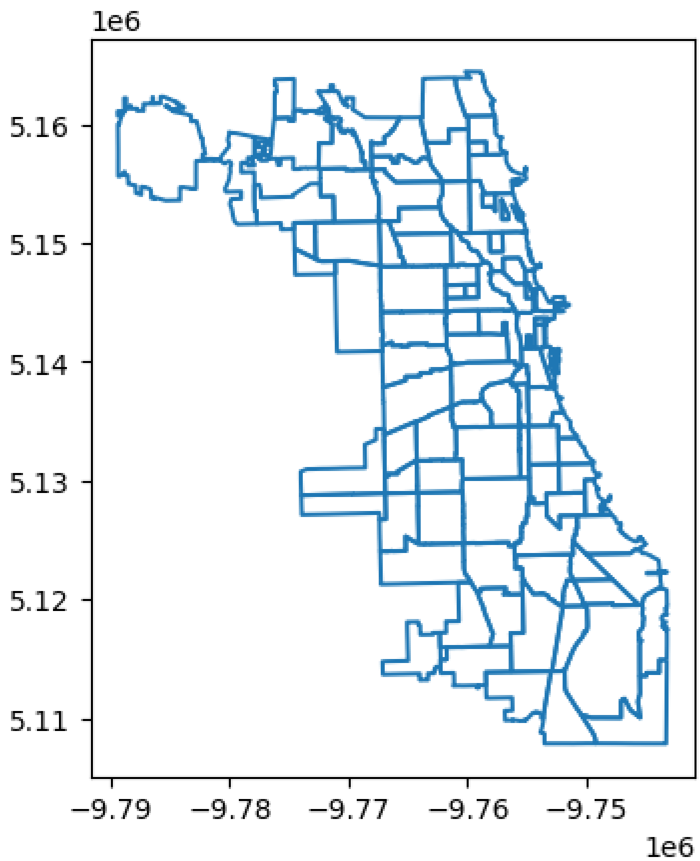
In [45]:
```
clawn = all_together[all_together['pri_neigh'] == 'Chicago Lawn']
clawn.plot()
```

Out[45]: <Axes: >



In [46]:
```
neighborhoods.boundary.plot()
```

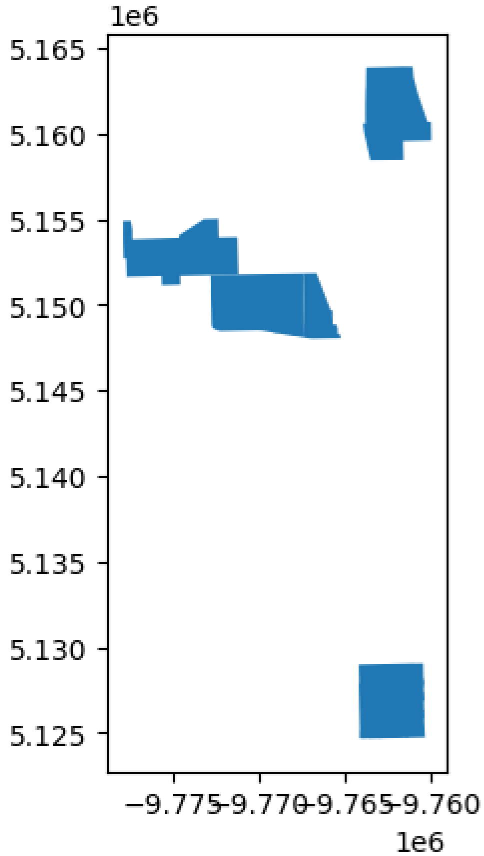
Out[46]: <Axes: >



In [47]:
```
low_fy = all_together[(all_together['pri_neigh'] == 'Hermosa') |
                       (all_together['pri_neigh'] == 'West Ridge')
                       | (all_together['pri_neigh'] == 'Belmont 
Cragin')
                       | (all_together['pri_neigh'] == 'Dunning')
                       | (all_together['pri_neigh'] == 'Chicago 
Lawn')]
```

In [48]:
```
low_fy.plot()
```

Out[48]: <Axes: >



In [ ]:
```
from shapely.ops import nearest_points

chi_tracts['centroid'] = chi_tracts.geometry.centroid
centroid = chi_tracts.set_geometry('centroid')

#sjoin_nearest
nearest = gpd.sjoin_nearest(centroid, l_stops,
distance_col='distance')
nearest
```

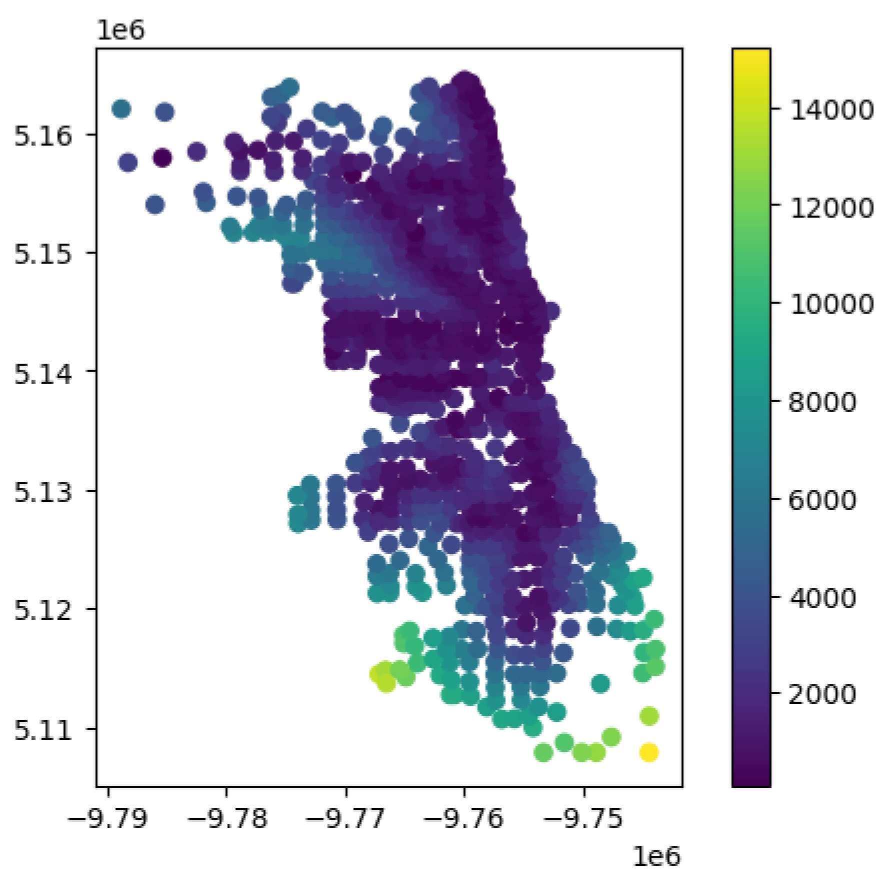
	OBJECTID	GEOIDFQ	ST	Name	Latitude
727	25419	1400000US17031640700	IL	6407	41.776412
722	25414	1400000US17031640100	IL	6401	41.781929
1480	26172	1400000US17031980100	IL	9801	41.785971
672	25364	1400000US17031560700	IL	5607	41.79646
1409	26101	1400000US17031835200	IL	8352	41.79664
...	...	...	...	...	...
163	24855	1400000US17031020301	IL	203.01	42.00691
1038	25730	1400000US17031808002	IL	8080.02	42.00507

	OBJECTID	GEOIDFQ	ST	Name	Latitude
1033	25725	1400000US17031807600	IL	8076	42.021442
161	24853	1400000US17031020100	IL	201	42.015751
1063	25755	1400000US17031810301	IL	8103.01	42.025327

868 rows × 75 columns

In [50]:
```
nearest.plot(column='distance', legend=True, cmap='viridis')
```

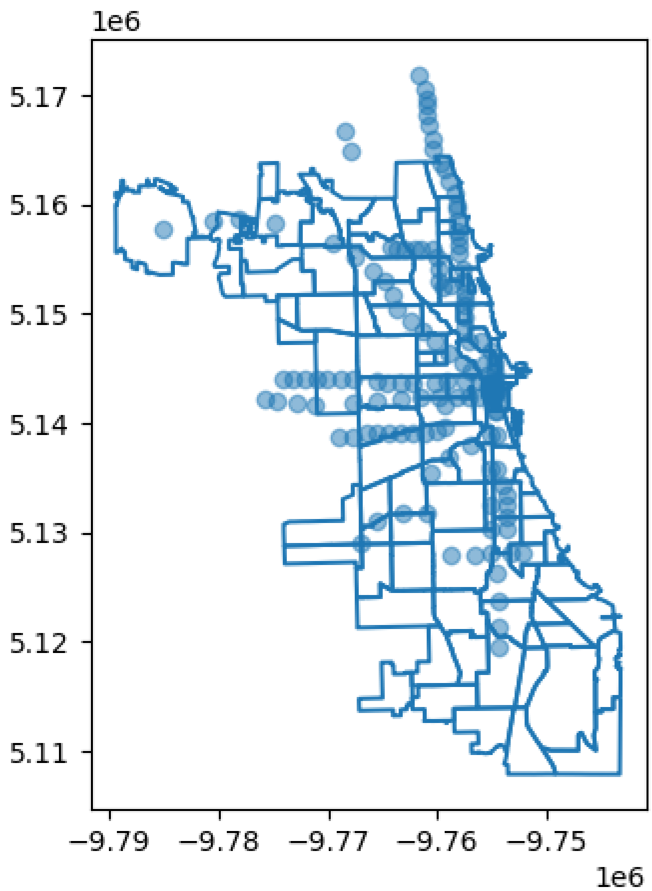
Out[50]: <Axes: >



In [51]:

```
fig, ax = plt.subplots(figsize=(10,5))

neighborhoods.boundary.plot(ax=ax)
l_stops.plot(ax=ax, alpha=0.5)
```

Out[51]: <Axes: >



In [52]:

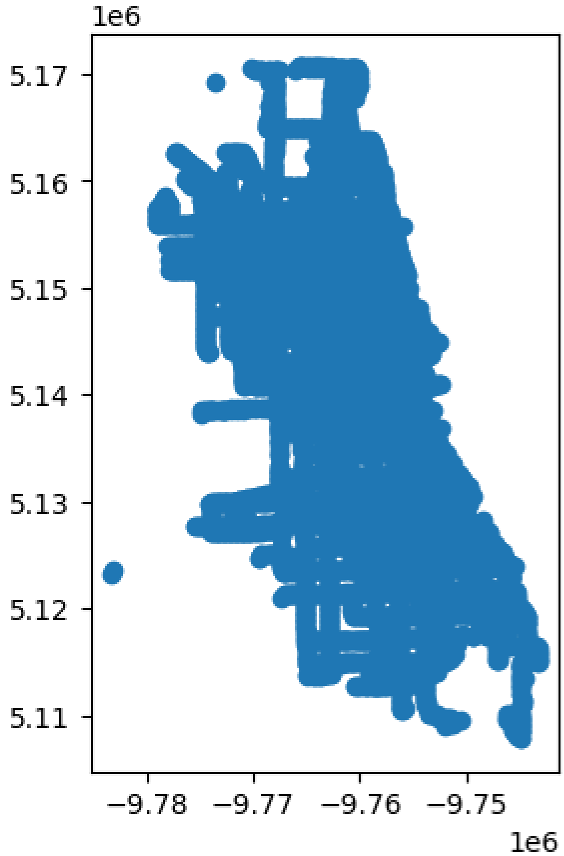
```
nearest.nsmallest(5, 'distance')
```

Out[52]:

	OBJECTID	GEOIDFQ	ST	Name	Latitude
392	25084	1400000US17031210601	IL	2106.01	41.937636
210	24902	1400000US17031040700	IL	407	41.966807
289	24981	1400000US17031081600	IL	816	41.892105
308	25000	1400000US17031110502	IL	1105.02	41.970278
499	25191	1400000US17031280100	IL	2801	41.885390

In [53]:
```
bus_stops.plot()
```

Out[53]: <Axes: >

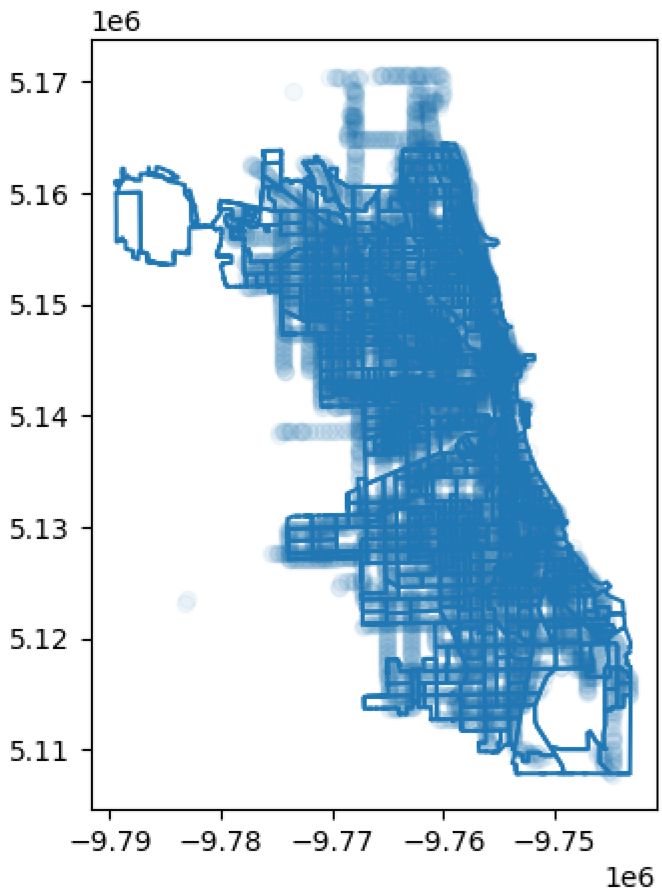


In [54]:
```
tract_bus = gpd.sjoin(chi_tracts, bus_stops, how='left', predicate='intersects')
```

In [55]:

```
fig, ax = plt.subplots(figsize=(10,5))

chi_tracts.boundary.plot(ax=ax)
bus_stops.plot(ax=ax, alpha=0.05)
```

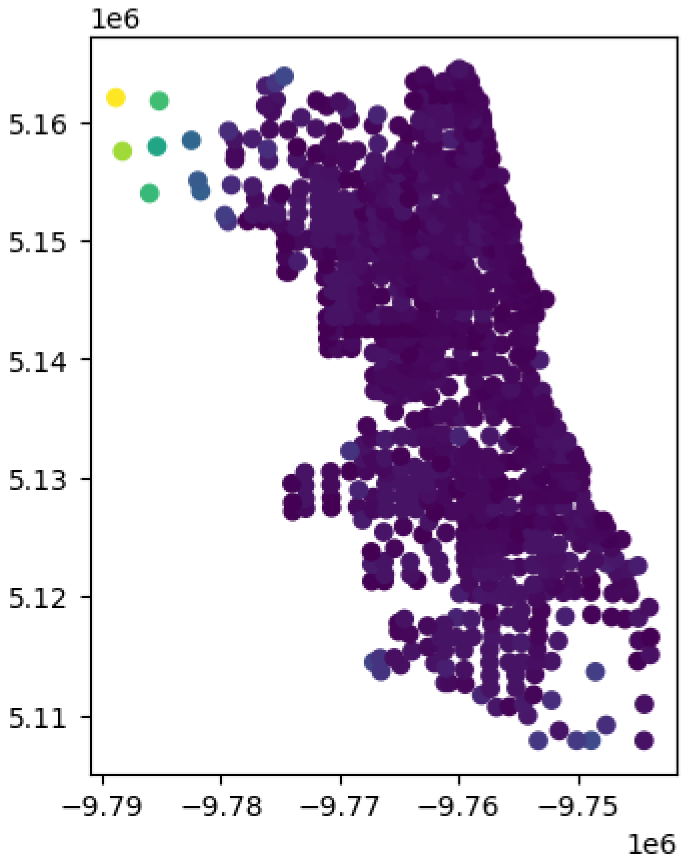
Out[55]: <Axes: >



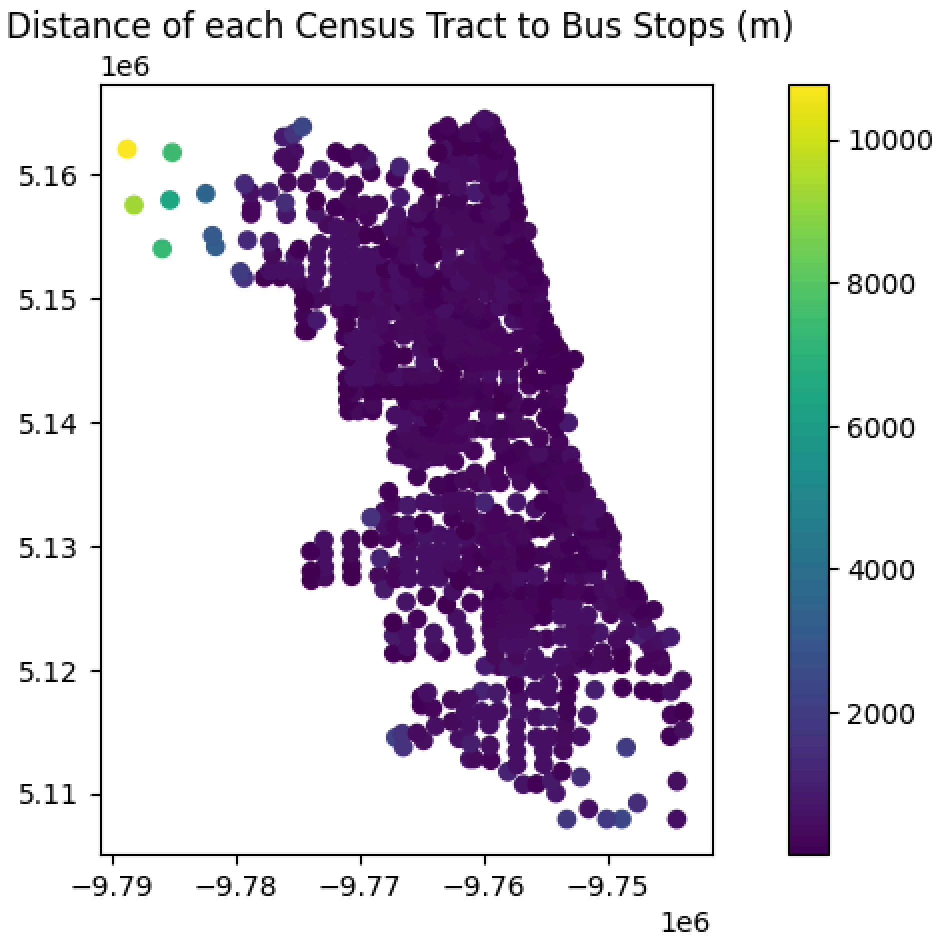
In [56]:

```
nearest_b = gpd.sjoin_nearest(centroid, bus_stops,
distance_col='distance')
nearest_b.plot(column='distance')
```

Out[56]: <Axes: >



In [57]:
```
fig,ax = plt.subplots(figsize=(10,5))

nearest_b.plot(column='distance', ax=ax, legend=True)
ax.set_title('Distance of each Census Tract to Bus Stops (m) ')
plt.show()
```

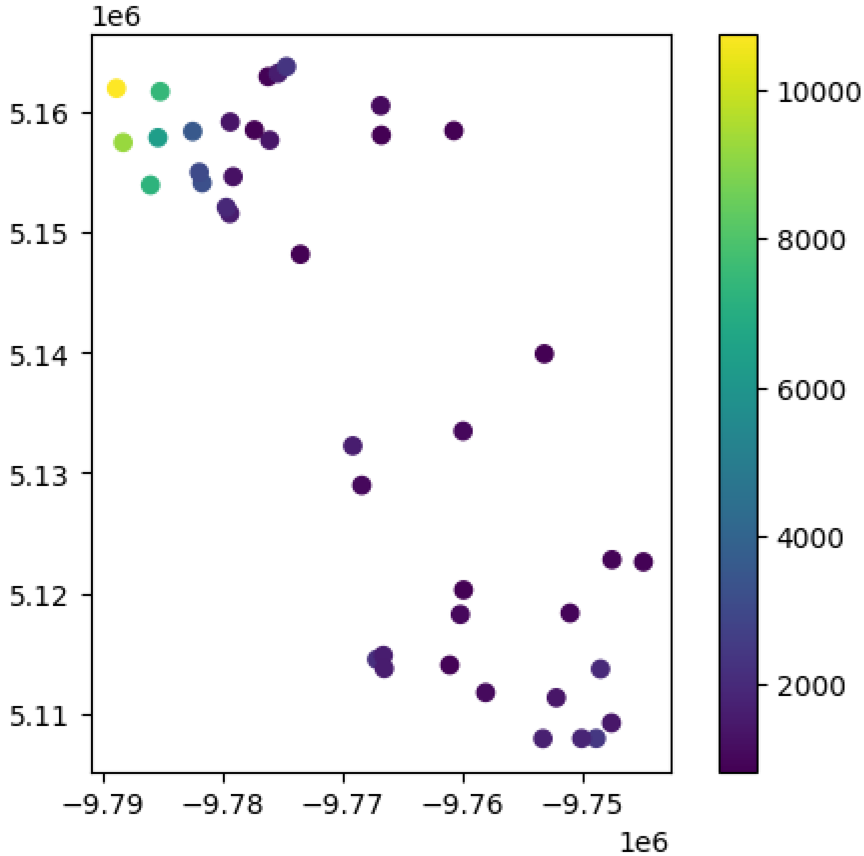


In [ ]:
```
# tracts with no stop within 0.5 (x > 800)
not_05 = nearest_b[nearest_b['distance'] > 800]
```

In [59]:

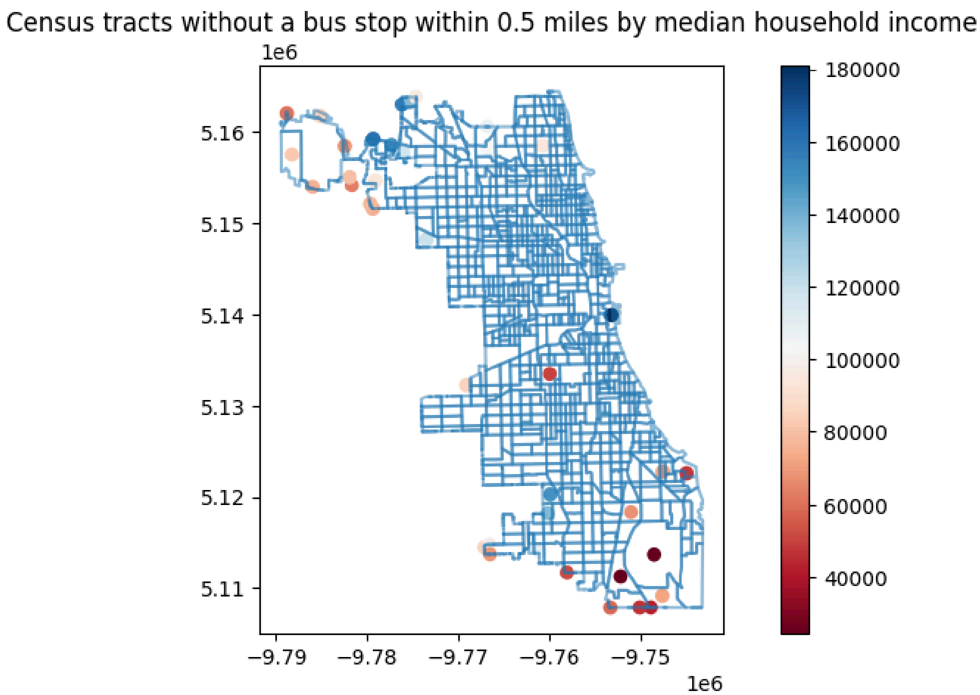
```
not_05.plot(column='distance', legend=True)
```

Out[59]: <Axes: >



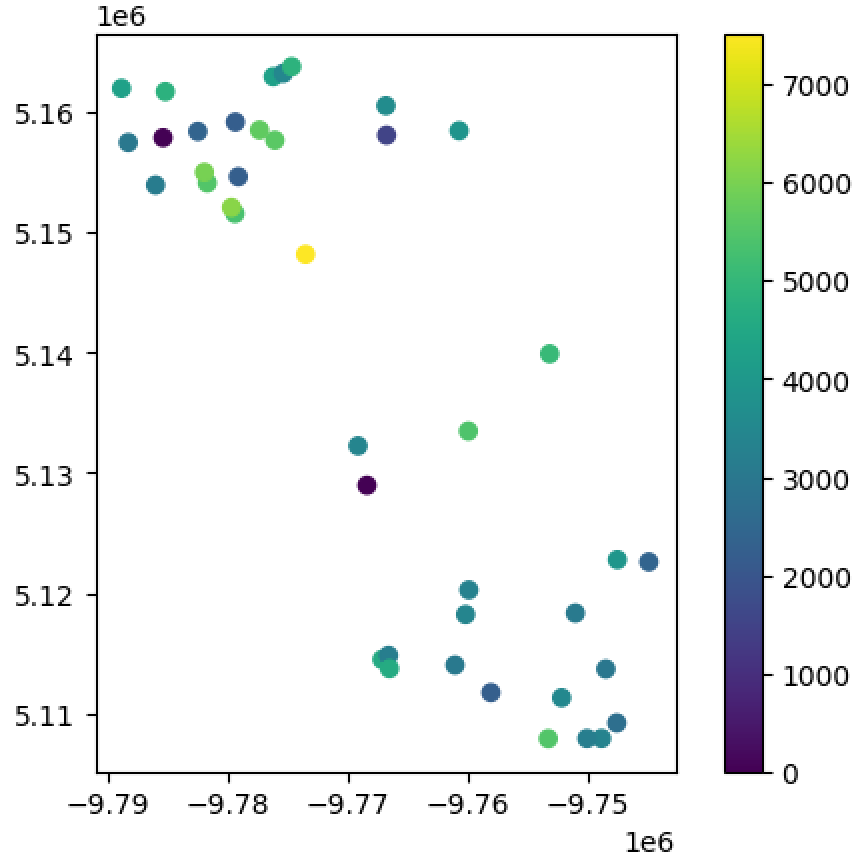
In [60]:
```
fig, ax= plt.subplots(figsize=(10,5))

not_05.plot(column='Median_Household_Income', ax=ax, legend=True,
cmap='RdBu')
chi_tracts.boundary.plot(ax=ax, alpha=0.5)
ax.set_title('Census tracts without a bus stop within 0.5 miles by
median household income')
```

Out[60]:
```
Text(0.5, 1.0, 'Census tracts without a bus stop within 0.5
miles by median household income')
```



In [61]:
```
not_05.plot(column='Total_Population', legend=True)
```

Out[61]: <Axes: >



In [62]:
```
print(not_05['Total_Population'].sum())
```

154428

In [63]:
```
chi_tracts['Total_Population'].sum()
```

Out[63]: 3056651

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